

A Guide to Best Practice for Electronic Materials Dispensing

**Alan Brewin, Ling Zou
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ABSTRACT

Robotic dispensing provides flexibility for electronics production not possible from screen-printing processes. As well as the application of solder paste and surface mount adhesive dispensing systems are utilised with newer technologies such as the application of conductive adhesives, chip on board globtops and flip chip underfills.

This document gives practical guidance for the dispensing of these materials for printed circuit assembly production.

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Approved on behalf of Managing Director, NPL, by Dr C Lea,
Head, Centre for Materials Measurement and Technology

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1 INTRODUCTION

Robotic dispensing is used for the application of a range of electronic production materials. Main examples include solder paste, conductive adhesives, surface mount adhesives, glob tops and underfills. This document provides practical guidelines for dispensing of these materials for those manufacturing using surface mount technology.

2 DISPENSING EQUIPMENT

Most automatic dispense systems have a servomotor driven Cartesian axis assembly to locate the dispensing nozzle assembly over the board, and a separate servomotor for positioning in the Z-axis.

Materials for dispensing are supplied by manufactures in plastic cartridges of varying sizes with EFD™ style lobes at one end for attachment of a pressure line, and Luer-lock™ thread at the other end for a nozzle arrangement to be fitted.

Often the robotic system will be able to accommodate different dispensing heads depending on requirements.

Machines are speed rated using Dots per Hour (DPH).

2.1 Pressure-Time Systems

Lower cost dispensing heads use pressure only to dispense material onto the PCB. The pneumatic syringe is normally fitted directly to a non-contact needle (See 2.6.2).

The greater the pressure and the longer it is applied for the more material that is dispensed. The repeatability and accuracy of these systems is less than for positive displacement systems (See 2.2), and dispensing smaller volumes are particularly problematic.

2.2 Positive Displacement Systems

Positive displacement systems use a mechanical device to move material through the nozzle, and pressure is only used to feed material into this mechanical pump.

2.2.1 Auger Screw Systems

An Auger screw (Figure 1) is a threaded spindle mounted in a cylindrical dispensing chamber, which is turned to force material out. Material is fed by low pressure into the side of the dispensing chamber, but it is the angle of rotation of the Auger screw that determines the volume of material dispensed, and its speed of rotation the material flow rate. An actuator lifts the screw upwards as it turns, and releases it down to plug the needle when it is motionless. Auger screw systems have much better accuracy and repeatability than pressure-time systems (See 2.1).

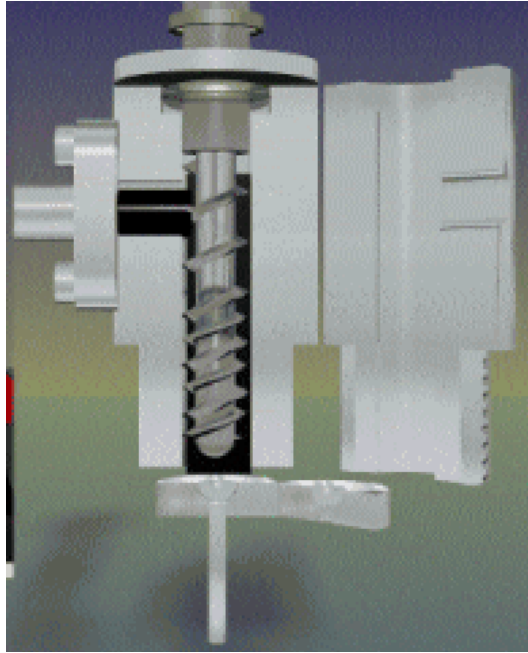


Figure 1. Schematic of an Auger Screw¹.

Data collected shows that the volume of material dispensed by Auger screw systems is directly proportional to the angle turned by the Auger screw.

By using screws with different pitches (angle of thread) different flow rates can be achieved. A coarser pitch Auger screw will produce a faster flow rate for a set speed of rotation, but will be slightly less accurate.

2.2.2 Piston Pump Systems

An arrangement of pistons can be used to dispense larger volumes of material (for encapsulation and potting). Single pump systems require a pause in dispensing while the piston is 'recharged' with material, but multi-piston systems now available allow continuous dispensing of larger volumes of material.

2.3 Cameras

Dispense systems should also have a camera system for locating the PCB using fiducials or other PCB landmarks. For accurate dispensing around components, further alignment may be necessary since components are not always exactly aligned with the board. An important part of the camera system is the edge detection software, which should be easily configurable so that all necessary components can be automatically found.

Problems can arise if contrast between the component and the laminate is low, or if edges are not well defined.

The system may also be able to check component orientation, or may look for 'bad marks' for chips with bad die.

2.4 Height Sensing

Whilst a camera system can locate PCB features in the X-Y plane it cannot determine height, (or position of the board surface in the Z-axis). Separate height

sensors are required for this purpose, unless a contact style of nozzle (See 2.6.1) is used.

Height sensing must often be performed many times across the area of a substrate as electronic laminates are often considerably warped.

2.4.1 Mechanical Touch Probe

A touch probe is lowered to the board surface at a desired location.

2.4.2 Laser Interferometer

A laser beam is bounced from the laminate surface and interferometry is used to determine its location. This method is faster than a touch probe and has the advantage of being non-contact.

2.5 Material Flow Calibration

The rheological properties of materials will often alter over time, for example as they cure. These changes can result in variation in flow rates and hence dispense volumes.

To counteract these effects some systems use a mass balance to relate the dispense time (valve open time) to the actual mass of material passing through the nozzle. Once the flow rate is determined in this way the system compensates dispense times to ensure the volume of material remains consistent. The required regularity of checking the mass depends on the material being used of course, but it should not be done too frequently as production speed is reduced.

An alternative approach uses material deposited onto a flat plate then a glass slide is pressed down onto the deposit to a known height. The size of the dot produced is measured by a camera system, and the volume of material present calculated.

Volume calibration systems have an advantage in that they are not affected by density changes of the material (for example filled materials can have a higher density at the bottom of a syringe due to settling).

2.6 Selection of Dispense Needles

Use a needle that is as short as possible, but will still allow access to the board substrate over the largest components. Shorter needles are less likely to become bent.

Dispensing relies upon the effects of surface tension to pull the material from the nozzle to the board surface. The surface tension holding the material to the nozzle should be small to ensure good transfer of material.

f = Surface tension between
needle and the product

F = Surface tension between
product and the surface

$F > f$ for good dispensing

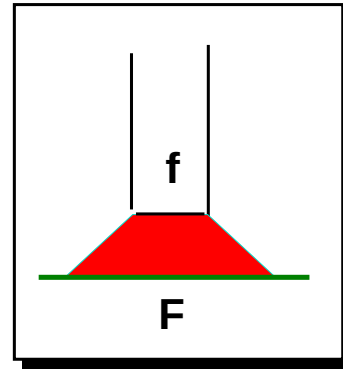


Figure 2. Surface tension forces during dispensing.

To help reduce the forces between the needle tip and the material chamfered needles can be used, and are recommended.

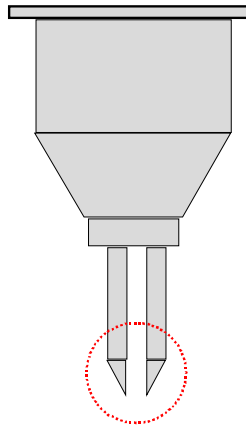


Figure 3. Chamfered needle will help reduce surface tension.

2.6.1 Contact (Footed) Needles

Contact nozzles are designed to come into contact with the board surface during dispensing. They have a solid 'foot' which is longer than the hollow nozzle through which the material flows and so are sometimes referred to as footed needles.



Figure 4. Footed Dispense Needle.

Since the foot is resting on the PCB substrate footed needles can only be used to dispense dots, not lines or other shapes. The foot determines the height of the needle above the substrate and no height sensing is necessary which saves considerable process time (See 2.4). Footed needles reduce the effect of board warping since height is determined locally for each dot dispensed.

It is important to bear in mind the position of the needle foot when programming a dispense pattern. The foot should not be placed into an area where there is a component, or where material has already been deposited. The foot will commonly be oriented at 45° to the needle to help avoid this.

The most common design for footed needles provides a dispense height twice the inner diameter of the needle. This arrangement is suitable for almost all dot-dispensing applications.

In order for a contact needle to be used the dispense head must hold the needle by means of a spring mounted assembly. This spring is compressed when the needle foot is pushed onto the PCA surface. Normally a compression of around 1-2mm can be used and this will accommodate any warping of the substrate.

2.6.2 Non-Contact Needles

When a line or other shape must be dispensed the needle cannot contact the PCB surface. Non-contact needles are simple in design (Figure 5), and for low accuracy applications disposable needles can be used (Figure 6).

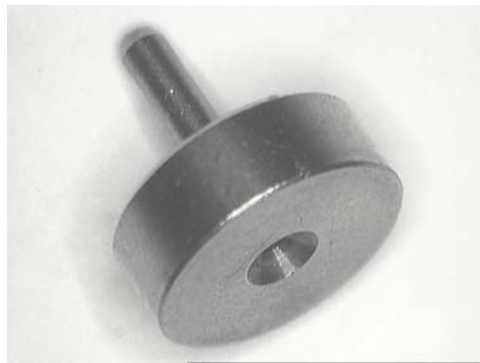


Figure 5. Non-contact needle.



Figure 6. Disposable non-contact needles, will screw directly onto material cartridges

2.7 Dispense Height

The distance between the substrate and needle tip is critical to the quality of dispensing. As a guide the height should be about 30 to 50% of the inner diameter of the chamfered needle.

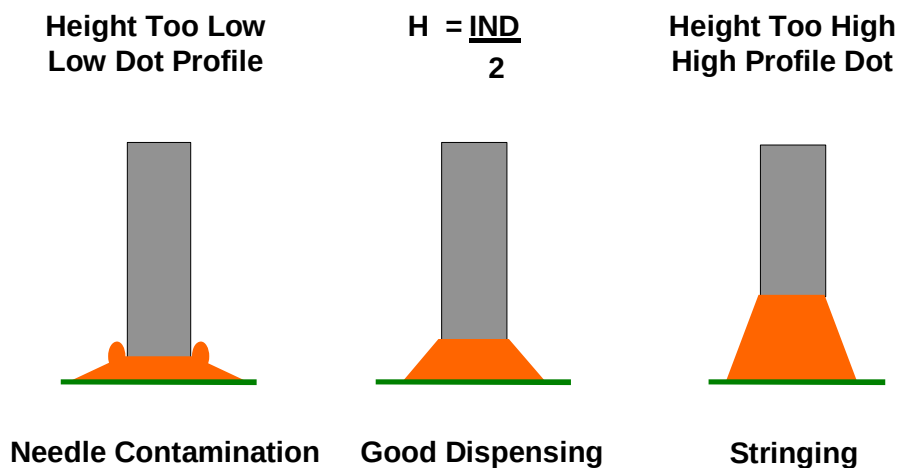


Figure 7. Choosing the correct dispense height

In practice a 2:1 ratio between needle inner diameter and dispense height gives good results for most materials.

2.8 PCB Handling

In-line systems will have automated handling of the PCB's. The industry standard protocol for conveyor and handling lines is SMEMA (Surface Mount Electronics Manufacturers Association). Ensure the equipment uses this protocol for smoother integration into your production.

The PCB will normally be carried to a mechanical end stop by the conveyor, although accurate positioning is not essential if the dispenser has a camera system. The PCB will then be firmly clamped on its edges.

It is important to ensure the board clamps are in line so that they do not induce a bend into the board (Figure 8).

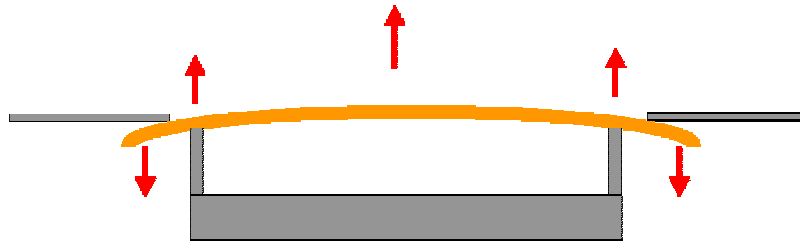


Figure 8. Bending of PCB due to poor clamping

The variation in height due to clamp bending can be much larger than that from inherent warping of laminates. With a footed needle, however, the practical dispense height will change over the board area due to bending. Figure 9 shows how variation in deposit shape may change over the PCB area if it is bent during mounting.

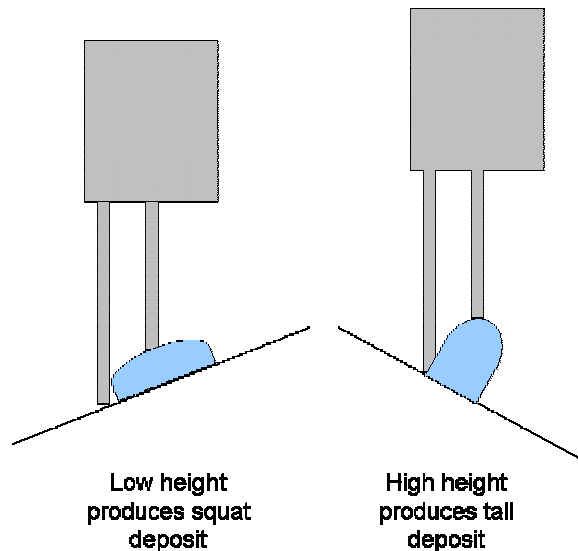


Figure 9. Effect of board bending on deposits from a footed needle

A practical example of this effect is shown in Figure 10. Profile scans of adhesive deposits dispensed with the same nozzle, and the same dispense time but at different board positions are shown.

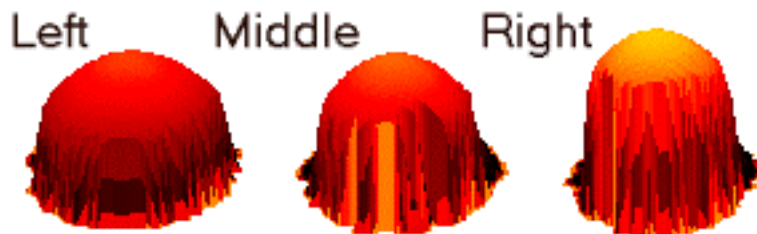


Figure 10. Scans of SM Adhesive deposits across a bent board

Non-contact needles are normally used for applications where lines are dispensed, and height is not so critical. Nevertheless be aware that a height probe also suffers from the same limitation as a footed needle if boards are bent by clamping.

2.9 Heating

2.9.1 Dispensing Head Heating

Ambient temperature variations can affect the rheology of materials as they sit in the machine mounted cartridge prior and during dispensing. These changes can lead to variation in dispensed volume even though all other parameters remain constant. Mass Calibration (See 2.5) can help reduce these effects, but often they can happen on a faster time scale than the relatively infrequent mass checks. To reduce the effects of ambient conditions some dispense head assemblies include a heater to ensure the material exits the nozzle at a constant temperature.

A typical set temperature for a heated nozzle would be between 30 and 40°C. This is sufficiently high above ambient but not too high to promote curing or setting of materials in the nozzle head.

Some materials performance is enhanced if it is dispensed at an elevated temperature.

Ambient humidity has little effect on dispense materials because they are contained within sealed cartridges.

2.9.2 Substrate Heating

Some dispensing systems have the option of substrate heaters and pre-heaters. Pre-heaters will heat the PCA from both sides using IR or radiation and convection. For heating in the actual dispensing area heating is only from the underside as the needle needs access to the top of the sample. Heating here is normally achieved by a hot plate, by radiation and convection.

Contact heating involves the vacuum attachment of a board specific heating plate to the underside of the board. This provides fast and even heating but is only suitable for high production volumes due to the inflexibility of tooling.

Mostly, it is only underfill materials (See 3.5) that require heated substrates during dispensing.

2.10 Cleaning

Machine suppliers will be able to provide specific guidelines on stripping down and cleaning dispensing heads. It should be noted, however, that cleaning of the complex mechanisms involved can be quite labour intensive and time consuming.

Equipment designed with strip-down and cleaning in mind can be much easier to clean, and this should be a consideration when looking to purchase dispensing heads.

Make sure to only use cleaning solvents recommended by the manufacturer and material supplier, and remember that if materials are left to cure in dispensing heads they may be rendered useless. An ultrasonic bath can help considerably with

cleaning of materials. Do not soak equipment in solvent overnight as any plastic parts, although resistant, may swell slightly.

2.11 Programming

Most modern dispense systems should utilise a user-friendly Windows™ style graphical interface. When investigating equipment make sure the efficiency, flexibility and ease of use of the software is suitable as well as the hardware of the dispenser itself. Long programming times, or mistakes in programmes can be costly.

The following parameters are commonly used and defined:

Dispense Height – Distance from needle tip to substrate during dispensing (defined mechanically by foot for contact needles, and variable within software for non-contact needles).

Up Height – Or lift height, this is the height the needle tip is lifted to after dispensing prior to any movement of the nozzle in the X-Y plane.

Dispense Time – Otherwise called valve-open time, it is the time over which material is dispensed from the needle.

Dwell Time – A pause after dispensing to allow wetting or sagging of material before the nozzle moves. Dwell times can prevent stringing, but lower DPH.

ON Delay – When dispensing a line a time delay can be used after the dispense valve is opened before the dispense head moves. This is to allow material to reach the board surface.

OFF Delay – When dispensing a line a distance off set can be used so that dispensing stops while the needle is still moving. This can prevent blobs, puddles or overshoot of material at the end of the line.

Note for use of Contact Needles (Figure 11): Be aware that the lift height as defined by the software as the distance the needle assembly is lifted in the Z-axis, NOT the height the needle tip is lifted to. In practice one must define the lift height in the software to be the required height the needle tip is lifted to PLUS the distance the spring mechanism is depressed during contact of the foot to the board.

$$\text{Software UH} = \text{Practical UH} + \text{Spring Deformation}$$

Any board warping will alter the amount of spring deformation, so a worst case Up Height should be used in the programming.

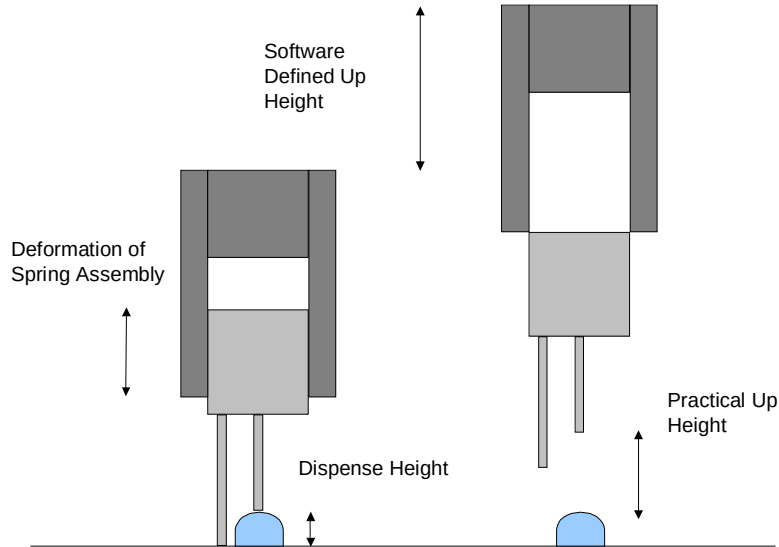


Figure 11. Practical and software defined Up Height when using a contact needle

One should also take care to ensure the program does not allow the foot of the contact needle to be placed into already deposited material.

3 DISPENSING SOLDER PASTE

Solder Paste consists of spherical particles of solder supported in a flux medium. Solder paste is applied to solderable PCB pads prior to the placing of electronic components. During reflow soldering the solder in paste melts and forms the solder joint connecting the component to the PCB.

Solder paste is applied by robotic dispensing for lower volume production where flexibility (not possible with screen printing) is important. For JIT (Just In Time) manufacturing the rapid product changeover and low downtime offered by dispensing is valuable. Throughput with dispensing will never compete with a screen-printing process.

Robotic dispensing of solder paste also finds uses in prototype production, applying paste to 3D laminates and automated rework.

Solder can be applied to give deposits of various heights; standard screen-printing will only produce deposits of a single height.

3.1 Needle Type

Solder paste is dispensed using dots only, and with a footed needle to maintain a high speed DPH.

A standard (foot height twice inner diameter) footed needle should be used. A Type 3 paste (25-45 μ m particle size) will be less prone to blocking needles, although smaller particle pastes can be more prone to oxidation in storage and may not wet as well in

reflow. Test results suggest that for a Type 2 paste you can use a 200µm ID needle, and for a Type 3 paste (45-75µm particles) a 300µm ID needle without blocking.

When blocking does occur with solder paste the flux can continue to be forced through the needle whilst particles are compressed together. When the nozzle head is disassembled to find the blockage, a 'bullet' can often be found and must be removed (Figure 12).

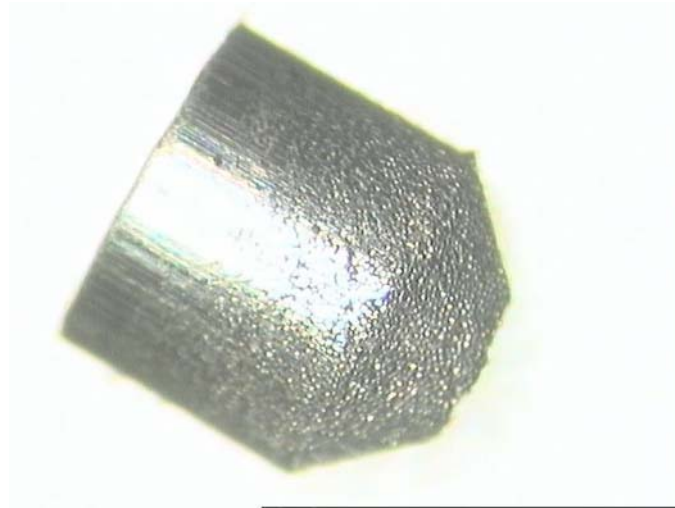


Figure 12. Compressed solder particles form a blockage

Pastes with the best dispensing performance have good homogeneity and spherical form of solder particles, and the metal content is well controlled.

Follow supplier guidelines unless you need to use a smaller needle for access. Using a larger needle will allow faster dispense rates for the material and help keep the speed of the process up.

Nozzle heating is normally not necessary unless ambient temperature is very variable.

3.2 Pump Type

A pressure-time system can be used to dispense solder paste, but variability in the dispensed volumes is possible. Pressure-time systems have particular limitation dispensing smaller dot sizes, and with valve open times of around 120ms or less, no paste is dispensed. The material absorbs energy every time the system is 'pumped', or pressurised to dispense a dot. This can result in heating of the paste, and also separation and cracking in the bulk of the material. The eventual result being missed dots.

Using an Auger screw system provides much better results and is recommended. As an example with a small ID needle (27 gauge), 100µg, 1.5mm paste dots can be achieved with a 10ms valve open time.

3.3 Stringing, Smearing and Satellites

Some pastes will form taller deposits than others for the same machine parameters due to their rheology. These products are more likely to string or smear.

If the *Up Height* of the needle is too low the needle will drag material across the board. This can happen because material is pushed ahead of the needle (Figure 13. Solder paste smearing) or because a string of material is drawn out between the deposit and the needle which then breaks and falls (Figure 14. Solder paste stringing).

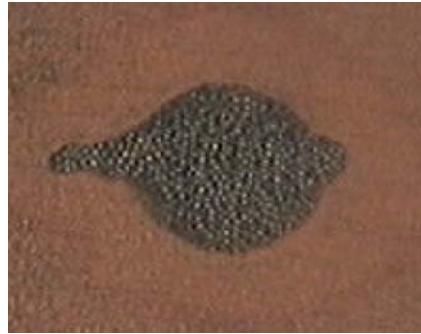


Figure 13. Solder paste smearing



Figure 14. Solder paste stringing

To avoid stringing or smearing the lift height must be increased so that the solder breaks contact with the needle before it moves across the board. An alternative is to introduce a dwell after deposition, but this is not very effective with solder pastes and reduces DPH more than lifting the nozzle slightly higher.

Sometimes small deposits of material are produced when stringing occurs. These are often randomly placed, and are referred to as satellites. Correcting the stringing problem will remove the satellite deposit too.

3.4 Dot Shape

For a given needle size dispensing a larger volume will produce a flatter paste deposit. Figure 15 is a plot showing how paste deposit diameter, height and volume change as dispense time increases (Auger screw dispense head). It is useful to note the following trends:

- Volume is directly proportional to dispense time
- Deposit diameter increases with volume, rate of increase reducing with volume
- The deposit height remains unchanged i.e. it is defined by the footed needle design

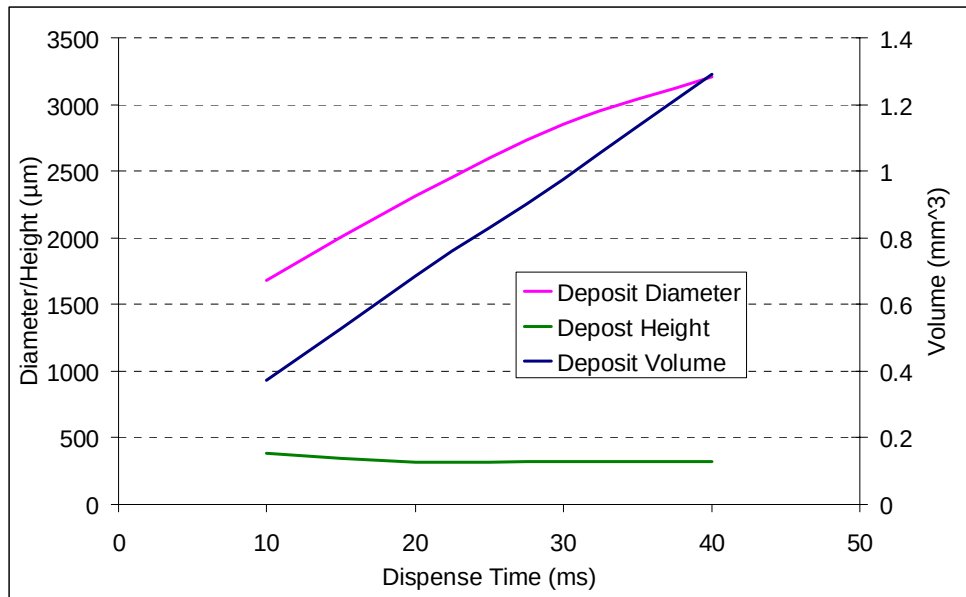


Figure 15. Changes in deposit height and diameter with increasing dispense time

This behaviour is in marked contrast to that of SM adhesive (Figure 26).

Figure 16 shows the actual edge profiles of an example paste as the dispense time is increased.

- To achieve higher, pointed deposits (for fine pitch work) use a larger needle (with a longer foot) but ensure you do not get missed dots (3.5).
- ??? axis label

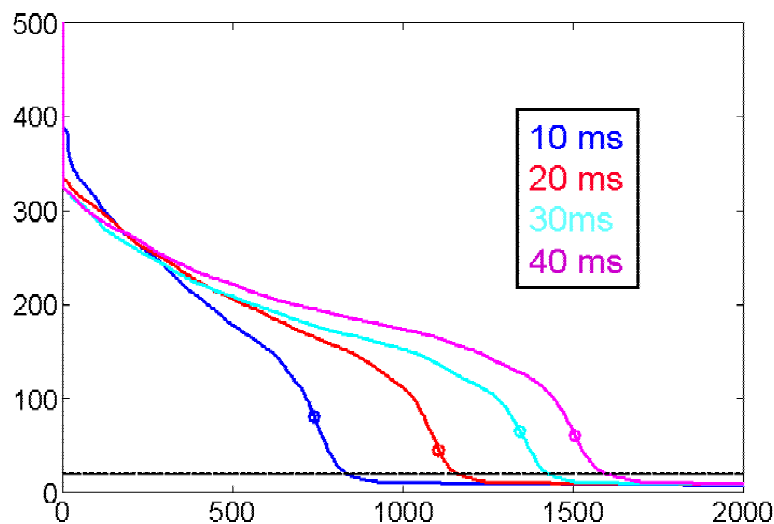


Figure 16. Deposit shape with increasing volume for increasing dispense time

3.5 Missing Dots and Big Dot Small Dot Defect

If the volume of paste dispensed from the needle is too small the paste may not make contact, or properly wet and adhere to the copper PCB pad. This results in a

dispensing defect known as a missed dot or big dot/small dot. The hallmark of this problem is a very small or missing dot, followed by a larger over-sized dot at the next deposition point, as illustrated in Figure 17.

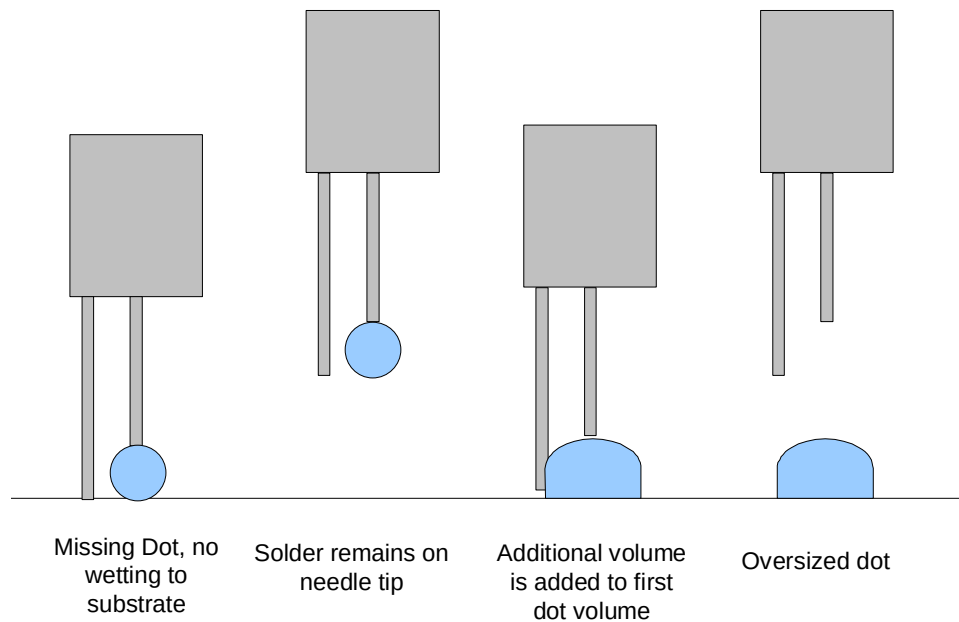


Figure 17. The 'missing dot' defect mechanism

The defect can be clearly seen in the array of deposits in Figure 18, where alternate dots are absent. Where there were missing dots, a flux circle can just be seen where contact is made.

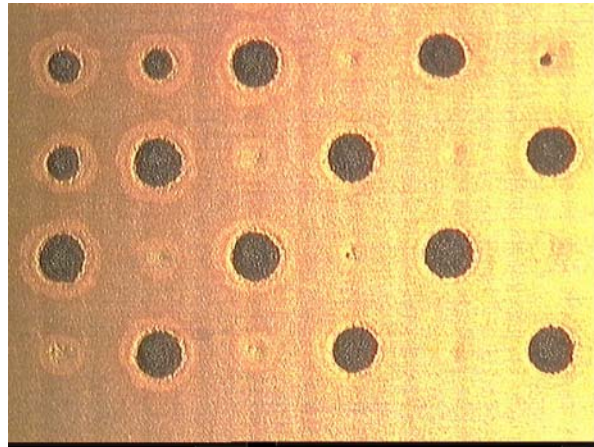


Figure 18. 'Missing dot' defect in an array of paste deposits

To eliminate the defect increase the dispense time, or if you want to keep the volume of paste down use a smaller ID and foot contact needle. The effect of changing the dispense time is shown in Figure 19.

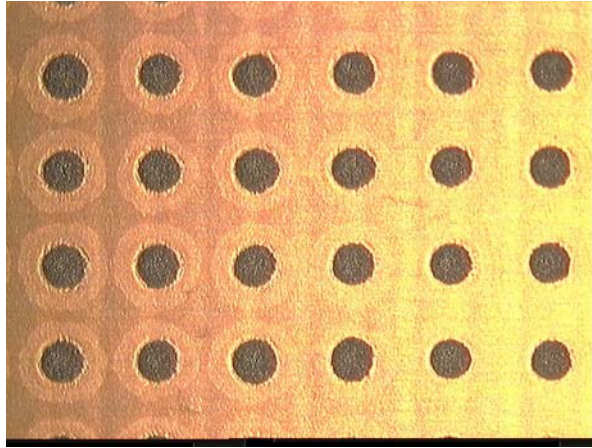


Figure 19. Elimination of 'missing dot' defect by increasing dispense time

3.6 Pad Layouts

For long or large pads a series of dots must be used to obtain the required volume of paste without depositing material outside the pad area. Solder paste deposited onto solder resist is a potential source of solder balling in reflow. Insufficient volume of paste, and component legs with poor planarity, may not contact the paste during placement, and no interconnection will be made.

Although dispensing many smaller dots can reduce overall process speed it is sometimes necessary to do this to avoid paste migrating from the pad area, as shown in Figure 20.

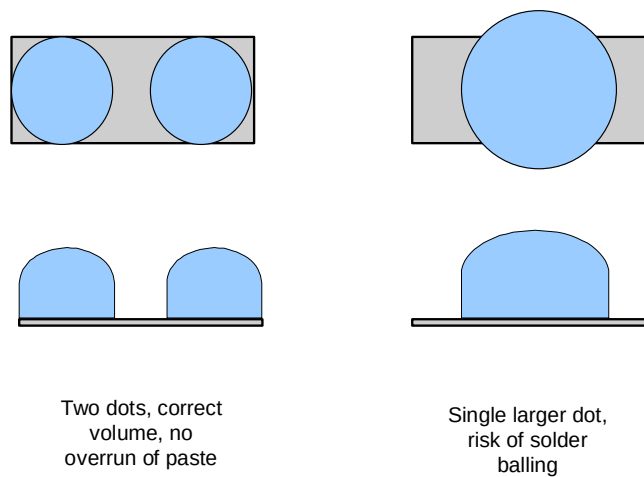


Figure 20. Smaller dots can achieve volume without overrun of paste

Figure 21 shows an example of the use of multiple dots deposited paste onto long QFP pads.

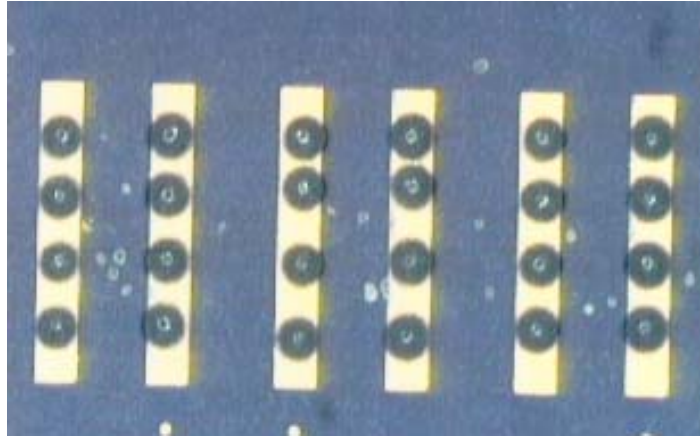


Figure 21. Multiple deposits used on long pads¹

For fine-pitch applications staggering of the deposits can help reduce the chance of 'wet shorts' or bridges of solder paste between the pads. These can form bridged joints during reflow. Figure 22 shows an example of staggering to achieve fine pitch dispensing.

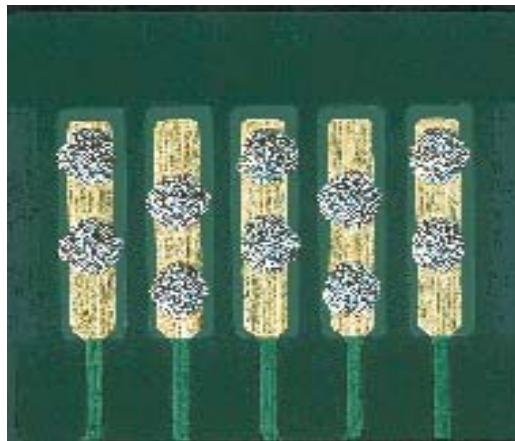


Figure 22. Staggering of deposits for fine pitch pads¹

4 DISPENSING SURFACE MOUNT ADHESIVE

Surface mount adhesive is applied, like solder paste, prior to component placement. It is positioned under the location of the component body, but not onto solderable pads. Components are placed and the adhesive is cured, its function being to hold component in place for subsequent wave solder operations.

Dispensing of adhesive is more common than screen application. This is because of the complication of having solder paste already deposited on the boards. Stencil printing is possible, but then an expensive relief stencil must be used, to avoid contact with the solder paste. This also provides higher adhesive.

4.1 Nozzle Type

SM adhesive is dispensed using dots only, and with a footed needle to maintain a high speed DPH.

A standard footed needle (foot height twice inner diameter) should be used. Generally blocking of nozzles is not an issue with SM adhesives, as they are not significantly loaded with particulates. A 200µm ID needle is recommended as a minimum, simply because thinner nozzles become more fragile.

Follow supplier guidelines unless you need to use a smaller needle for access. Using a larger needle will allow faster dispense rates for the material and help keep the speed of the process up.

Different SM adhesive products have characteristic deposit shapes (some more tall and conical, some more dome shaped) but there is no specific advantage to any shape, as long as the deposit height is sufficient to contact the component body. Less material can be needed if your deposit shape is tall and thin.

Nozzle heating is normally not necessary unless ambient temperature is very variable.

4.2 Pump Type

A pressure-time system can be used to dispense SM adhesives, but expect variability in the volumes dispensed. Pressure-time systems have a particular limitation in dispensing smaller dot sizes, and with valve open times of around 120ms or less, no paste is dispensed. The material absorbs energy every time the system is 'pumped' with pressured air to dispense a dot. This can result in heating of the adhesive, reducing its pot life.

Using an Auger screw system provides much better results and is recommended. As an example with a small ID needle (27 gauge), 110µg, 1.2mm diameter adhesive deposits can be achieved with a 10ms valve open time.

4.3 Stringing, Smearing and Satellites

Some adhesives will form taller deposits than others for the same machine parameters due to their rheology. These are the products that are more likely to string or smear.

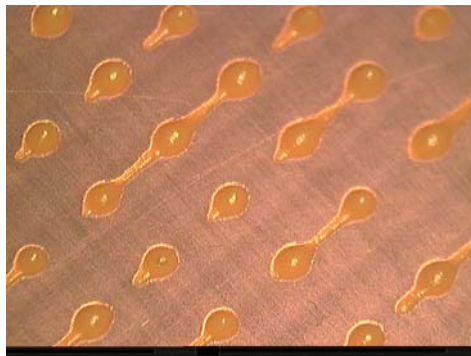


Figure 23. SM Adhesive Smearing

If the *Up Height* of the needle is too low the needle will drag material across the board. This can happen because material is pushed along with the needle, shown in Figure 23, or because a string of material is drawn out between the deposit and the needle that then breaks and falls, demonstrated in Figure 24.



Figure 24. SM Adhesive Stringing

To avoid stringing or smearing the lift height must be increased so that the adhesive breaks contact with the needle before it moves across the board. An alternative is to introduce a dwell after deposition. Adhesives tend to be more prone to stringing than solder pastes due to their higher cohesive strength, and the partition of material from nozzle happens at higher height.

Sometimes additional small deposits of material are produced when stringing occurs. These are often randomly placed, and are referred to as satellites. Correcting the stringing problem will remove the satellite deposits too.

4.4 Dot Shape

For a given needle size dispensing a larger volume of adhesive will produce a taller, thinner adhesive deposit. Figure 25 is a plot showing how SM adhesive deposit diameter, height and volume change as dispense time increases (Auger screw dispense head). It is useful to note the following trends:

- Volume is directly proportional to dispense time
- Deposit radius increases with volume, but this rate of increase reduces with increasing volume
- The deposit height also increases with volume.

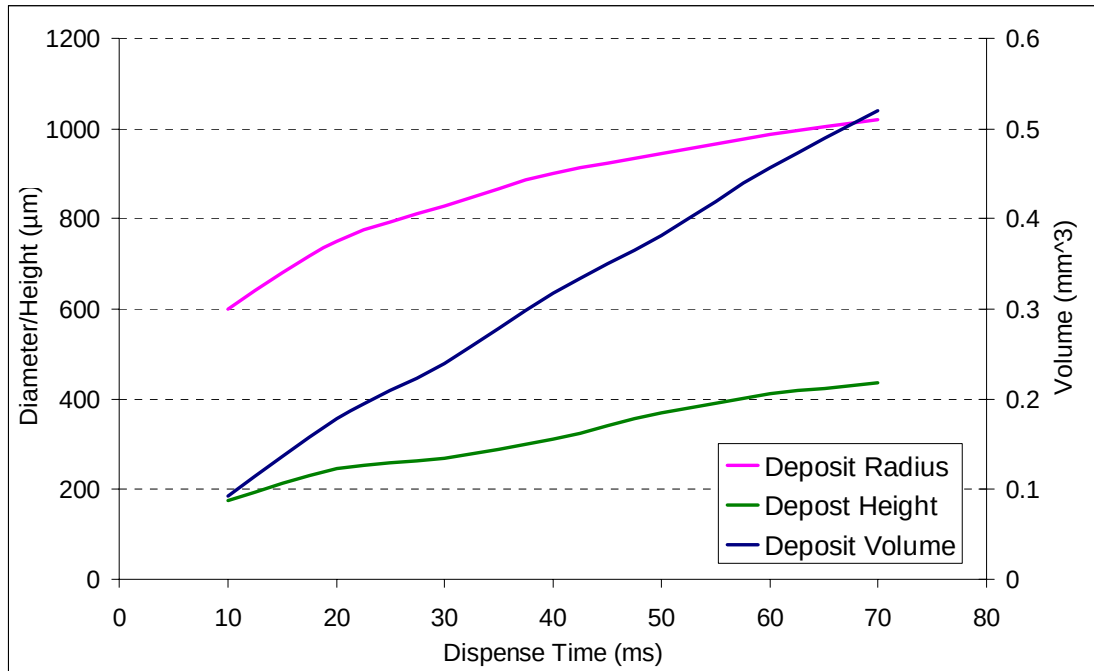


Figure 25. Changes in deposit height and diameter with increasing dispense time

It is useful to highlight the different behaviours of solder pastes and adhesives. This is shown graphically in Figure 26 where the height/radius ratio is plotted against dispensed volume for two sample materials. A high height/radius ratio means a tall, thin deposit. For the adhesive the deposit becomes taller and thinner in shape.

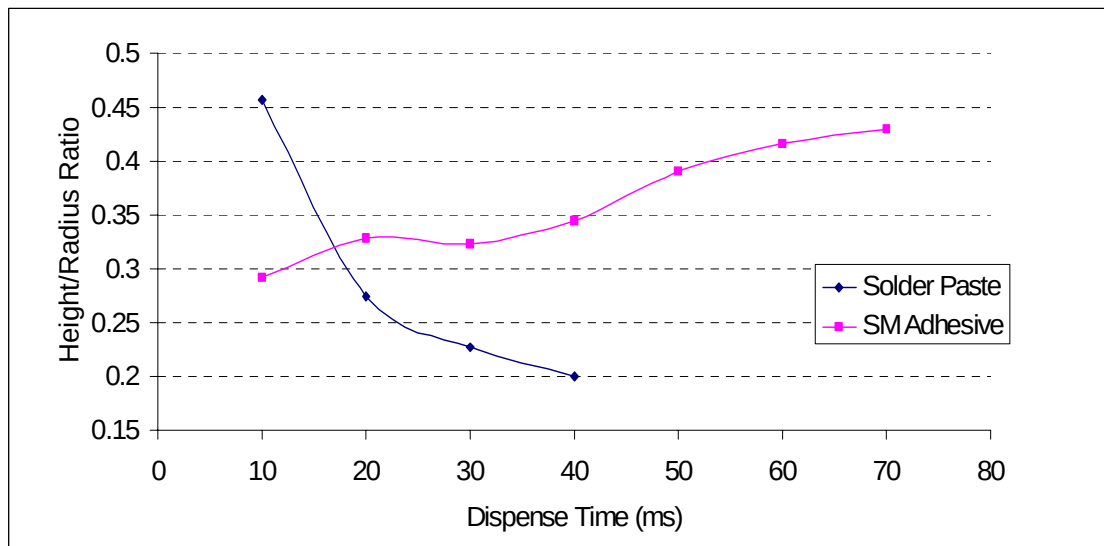


Figure 26. Changes in height/radius ratio for solder paste and adhesives

Figure 27 shows the actual edge profiles of an example SM adhesive as the dispense time is increased.

- To achieve higher, pointed deposits (for smaller SM components) use a smaller needle (with a shorter foot).

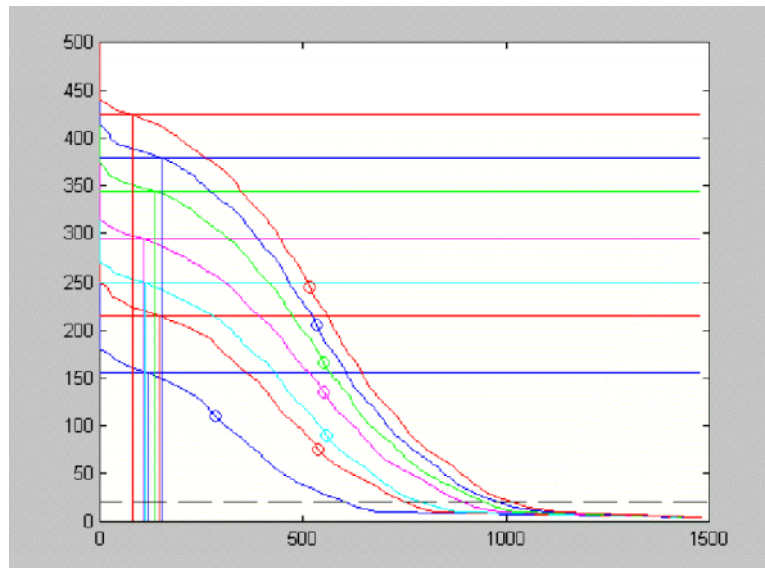


Figure 27. Deposit shape with increasing dispense time (10,20,30,40,50,60,70ms)

4.5 Missing Dots, and, Big Dot Small Dot Defect

If the volume of adhesive dispensed from the needle is too small it may not make contact, or properly wet and so adhere to the copper PCB pad. This results in a dispensing defect known as a missed dot, or big dot/small dot, as was discussed in section 3.5. The hallmark of this problem is a very small or missing dot, followed by a larger over-sized dot at the next deposition point, shown previously in Figure 17.

The defect can be clearly seen in the array of SM adhesive deposits in Figure 28, where alternate dots are very small. The dots were dispensed in rows from left to right.

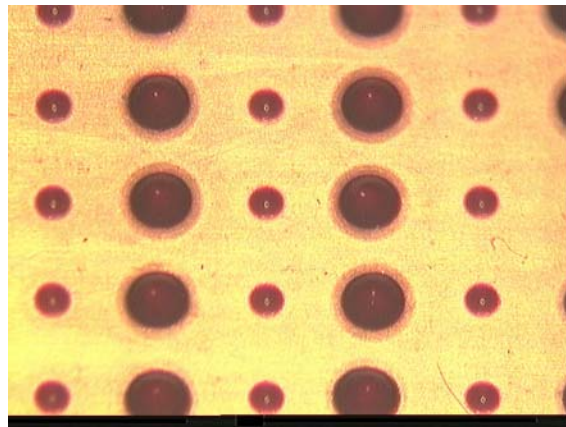


Figure 28. 'Missing dot' defect in an array of SM adhesive deposits

To eliminate the defect increase the dispense time, as shown in Figure 29, or if you want to keep the volume of adhesive down use a smaller ID and foot contact needle.

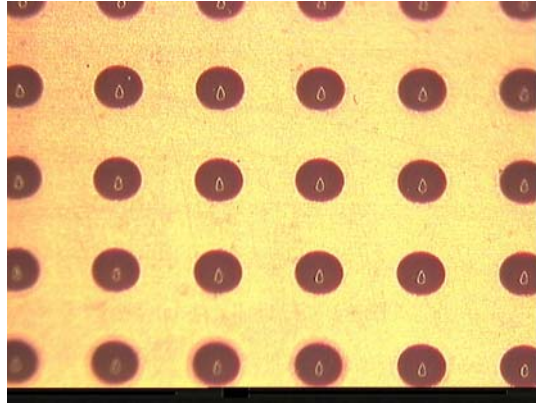


Figure 29. Elimination of 'missing dot' defect by increasing dispense time

4.6 Deposit Layouts

The SM adhesive is only required to hold the component in place through the wave soldering process; the solder is designed to give mechanical performance longer term. As such a minimum amount of adhesive should be used to contact the component underside, without adhesive spreading onto pad/paste areas.

When calculating the require height and volume of dot remember to include the thickness of the copper pads (Figure 30).

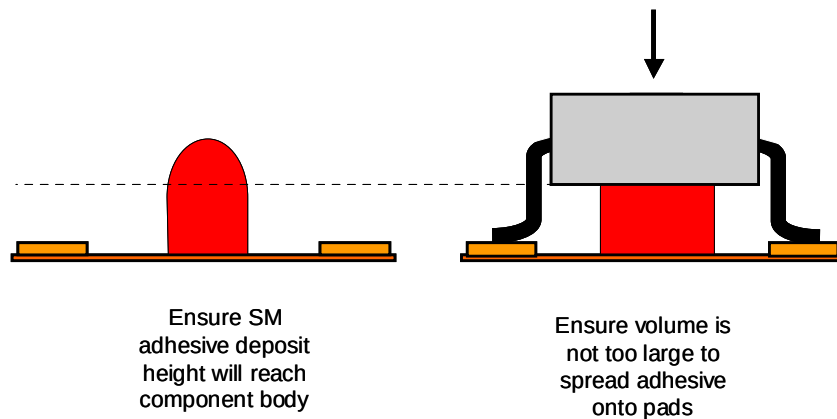


Figure 30. SM adhesive deposit size

For very small SM components it can sometimes help to place adhesive dots near the component body perimeter (Figure 31). When the component is pushed into the adhesive it is forced outwards away from the component and spreads less. This technique can avoid adhesive spreading onto the copper pads.

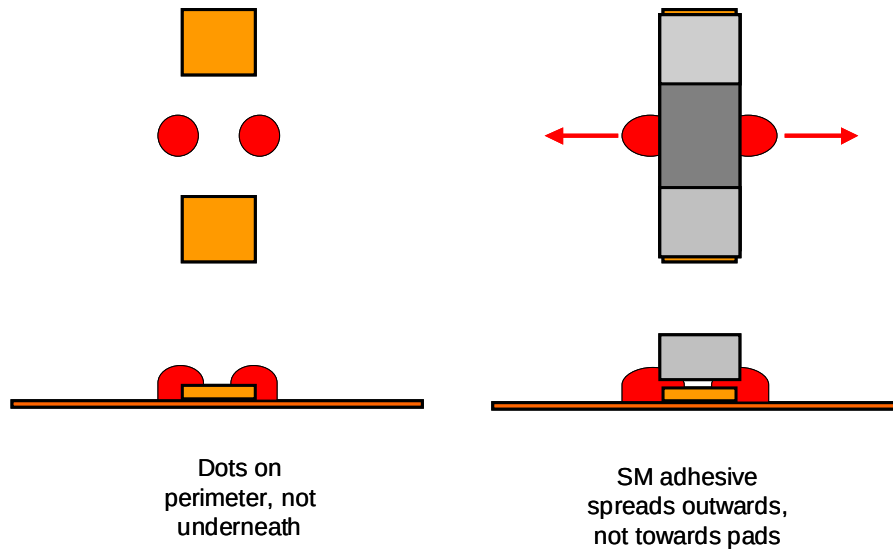


Figure 31. SM adhesive applied at component perimeter

5 DISPENSING ISOTROPIC CONDUCTIVE ADHESIVE

Isotropic conductive adhesive (ICA) consists of electrically conductive silver flakes (approximately 80% by mass) supported in an epoxy adhesive medium. ICA is applied to PCB pads prior to the placing of electronic components. During subsequent curing the adhesive cures and forms the conductive joint connecting the component to the PCB.

ICA is applied by robotic dispensing for lower volume production where flexibility (not possible with screen printing) is important. For JIT (Just In Time) manufacturing the rapid product changeover and low downtime is offered by dispensing in valuable. Throughput with dispensing will never compete with a screen-printing process, however.

5.1 Needle Type

ICA is dispensed using dots only, and with a footed needle to maintain a high speed DPH.

A standard (foot height twice inner diameter) footed needle should be used. Test results suggest that for a typical ICA you can use a 200µm ID needle without blocking.

Follow supplier guidelines unless you need to use a smaller needle for access. Using a larger needle will allow faster dispense rates for the material and help keep the speed of the process up.

Nozzle heating is normally not necessary unless ambient temperature is very variable.

5.2 Pump Type

A pressure-time system can be used to dispense ICA, but expect variability in the volumes dispensed. Again pressure-time systems have a particular limitation dispensing smaller dot sizes. The material absorbs energy every time the system is 'pumped' with pressured air to dispense a dot. This can result in heating of the adhesive, and also separation and cracking in the bulk of the material. The eventual result being missed dots.

Using an Auger screw system provides much better results and is recommended. As an example with a small ID needle (27 gauge), 240 μ g, 0.6mm ICA dots can be achieved with a 10ms valve open time.

5.3 Stringing, Smearing and Satellites

Some ICA will form taller deposits than others for the same machine parameters due to their rheology. These are the products that are more likely to string or smear.

If the *Up Height* of the needle is too low the needle will drag material across the board. This can happen because material is pushed ahead of the needle or because a string of material is drawn out between the deposit and the needle which then breaks and falls.

To avoid stringing or smearing the lift height must be increased so that the ICA breaks contact with the needle before it moves across the board, see Figure 32. An alternative is to introduce a dwell after deposition, but this is not very effective with ICAs and reduces DPH more then lifting the nozzle slightly higher.

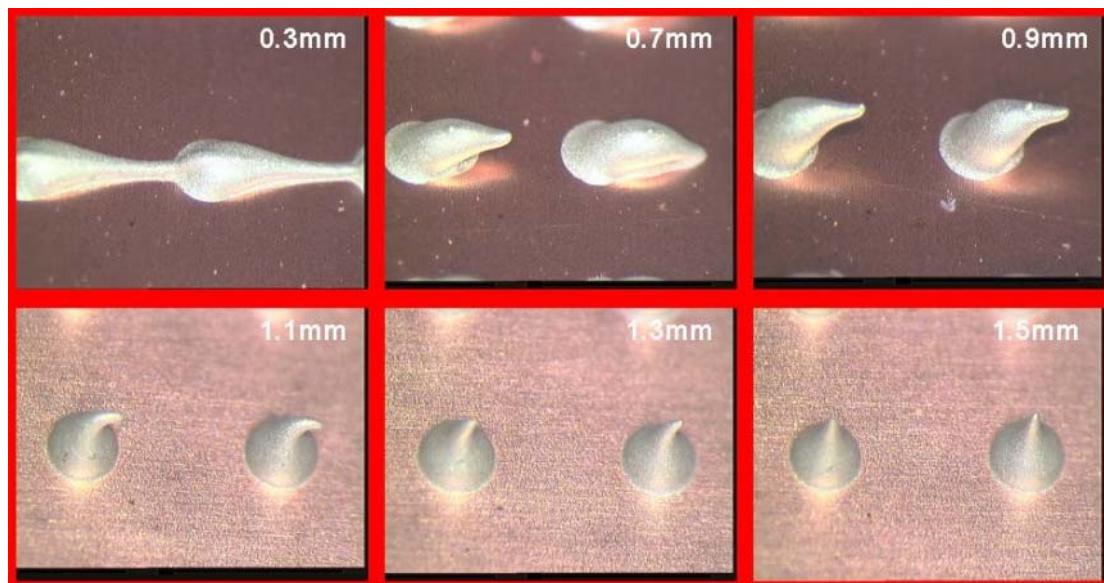


Figure 32. ICA Smearing and stringing eliminated by increasing needle lift height

Sometimes small deposits of material are produced when stringing occurs. These are often randomly placed, and are referred to as satellites. Correcting the stringing problem will remove the satellite deposit too.

5.4 Dot Shape

For a given needle size dispensing a larger volume will produce a flatter ICA deposit. Figure 33 plots how paste deposit diameter, height and volume change as dispense time increases (Auger screw dispense head). It is useful to note the following trends:

- Volume is directly proportional to dispense time
- Deposit radius increases with volume, the rate of increase being greater than for height
- The deposit height increases with volume

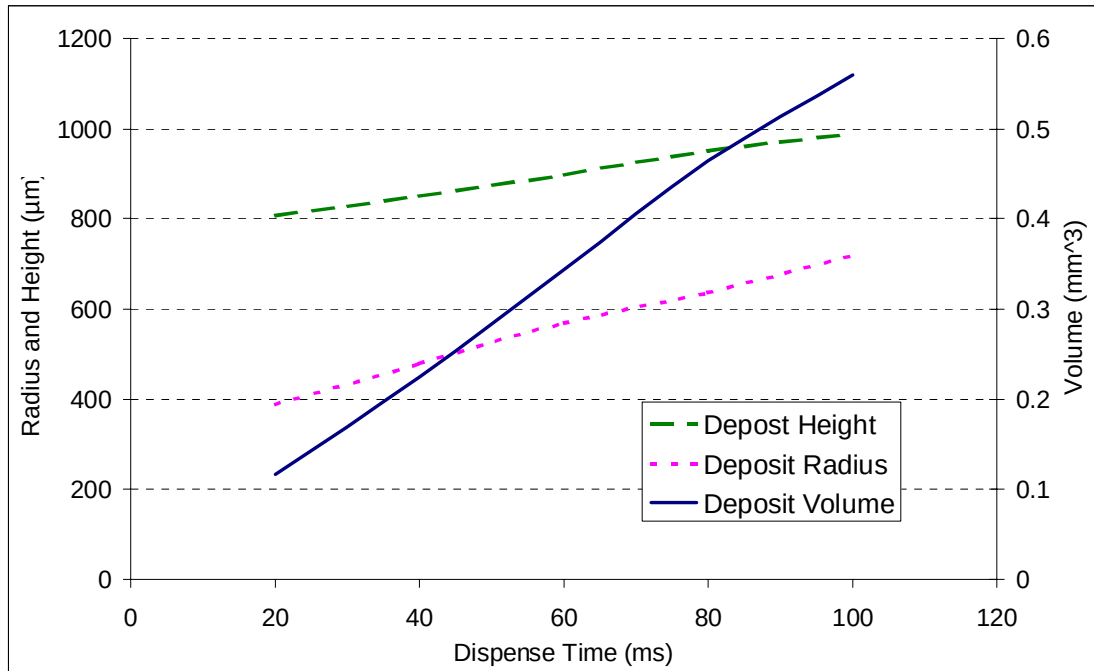


Figure 33. ICA deposit height, radius and volume with increasing dispense time

It is useful to highlight the different behaviours of solder pastes, SM adhesives, and ICAs. This is shown graphically in Figure 26 where the height/radius ratio is plotted against dispensed volume for three sample materials. A high height/radius ratio means a tall, thin deposit.

The behaviour of the ICA is in marked contrast to that of SM adhesive but similar to that of solder paste.

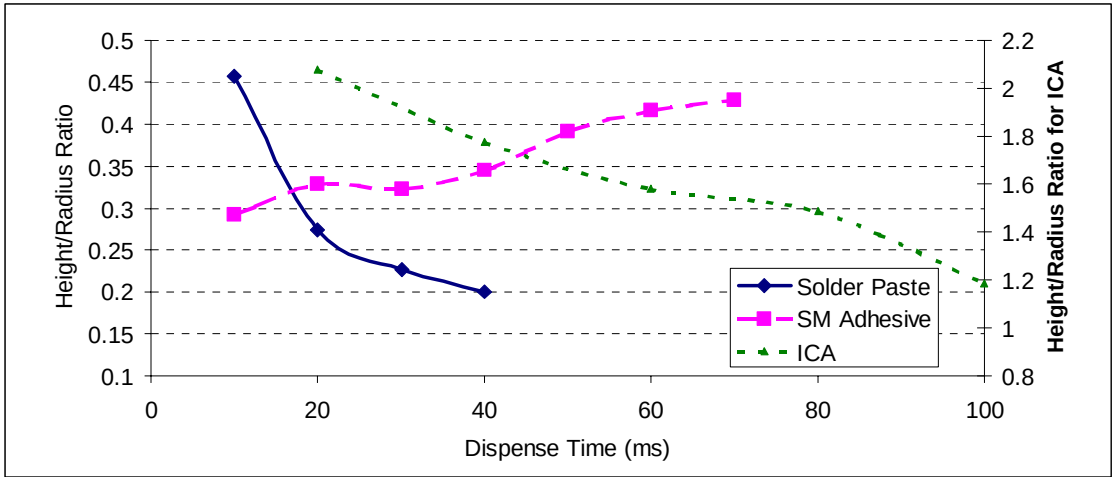


Figure 34. Changes in height/radius ratio for solder paste, adhesive and ICA

Figure 35 shows the actual edge profiles of an example ICA as the dispense time is increased.

- To achieve higher, pointed deposits (for fine pitch work) use a larger needle (with a longer foot) but ensure you do not get missed dots (3.5).

???? annotate figure and increase axis labels

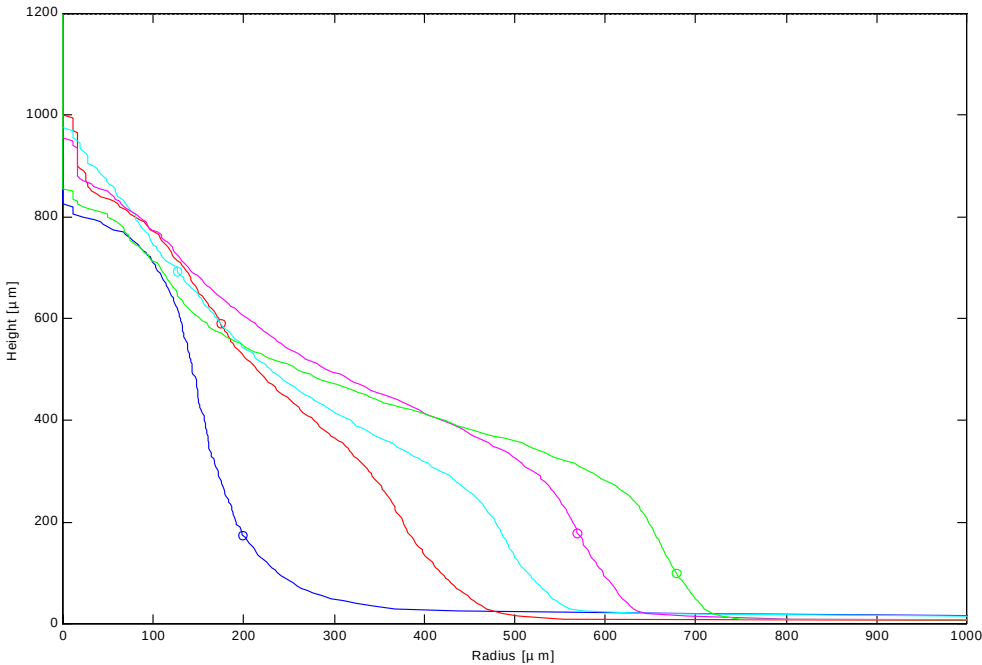


Figure 35. ICA deposit shape with increasing volume

5.5 Missing Dots and Big Dot Small Dot Defect

This is very similar to the issues encountered previously with solder paste and SM adhesive, as discussed in sections 3.5 and 4.5.

To eliminate the defect increase the dispense time, or if you want to keep the volume of adhesive down use a smaller ID and foot contact needle. Examples of this defect can be seen in Figure 18, Figure 19, Figure 28 and Figure 29.

5.6 Pad Layouts

Considerations for pad layout are the same as for solder paste (see 3.6). Additionally the volume and height of ICA required to form a good joint is likely to be greater than that experienced with solder paste. ICA does not wet up the side of component legs during cure, unlike paste that does during reflow (Figure 36). The ICA is displaced during component placement and the only material in contact with the component metallisation at this point, will result in a conductive path.

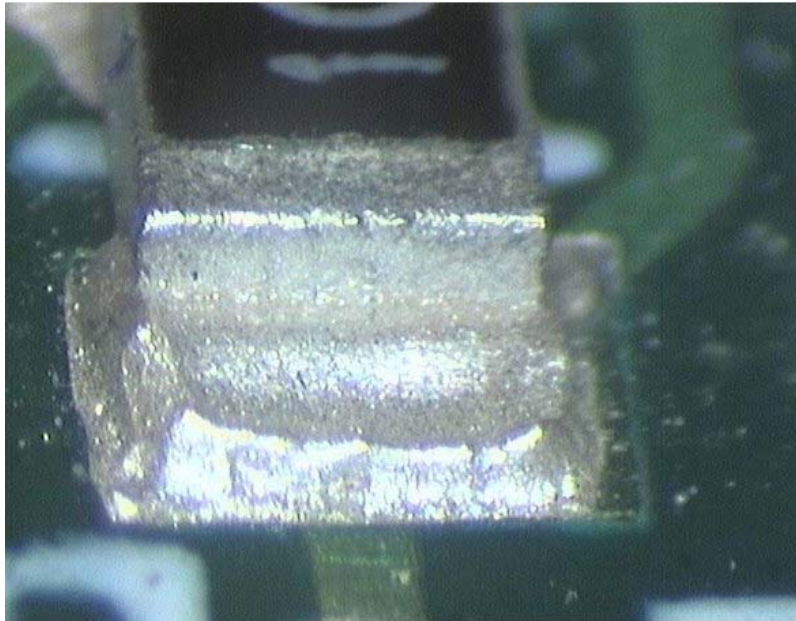


Figure 36. Typical cured ICA joint with no wetting.³

6 DISPENSING GLOB TOPS

Glob tops are encapsulation materials applied over Chip on Board (COB) components to protect them from the environment and provide mechanical support. In effect glob tops are the component body for COB technology components.

Simple glob top materials are one-part systems, which are applied over the chip area. 'Dam and Fill' glob tops consist of two components. A high viscosity dam material is applied around the chip, and then a low viscosity fill material is used to fill in the dammed area.

6.1 Using an ON-delay to dispense a line

When material is forced from a needle it can take hundreds of milliseconds for the material to actually reach the PCB surface. This effect means that if the dispense equipment moves the needle to begin dispensing a line, and opens the dispense valve at the same time, the material can actually be deposited some distance from the programmed start point.

To overcome this an ON-Delay can be used. This is a time delay between the opening of the dispense valve, and the movement of the dispense needle.

Figure 37 shows the effect of changing the ON-delay setting. The time required is dependant on the material, the nozzle diameter, the dispense height and the dispense speed. It must be found by trial and error for any particular set up.

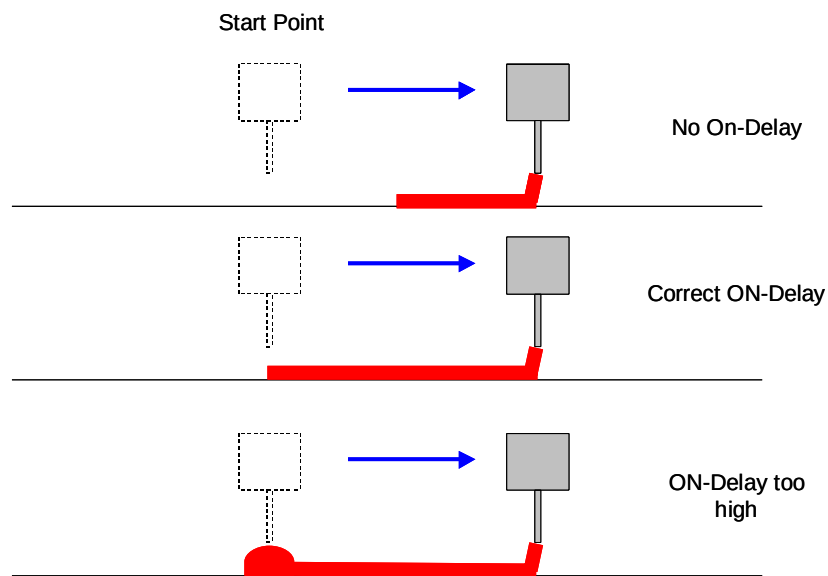


Figure 37. Using an ON-Delay to start a dispense a line

6.2 Using an OFF-Delay for line dispensing

When material flow from a needle is halted by closing the dispense valve, material still below the valve can continue to flow out onto the PCB. This effect means that if the dispense equipment stops and lifts the dispense needle at the end of line, closing the dispense valve at the same time, additional material can actually be deposited at the end point (i.e. pooling).

To overcome this an *OFF-Delay* can be used. This is defined as the distance before the end of line position at which the dispense valve will be closed.

Figure 38 shows the effect of changing the *OFF-delay* setting. The time required is dependant on the material, the nozzle diameter, the dispense height and the dispense speed. It must be found by trial and error for any particular set up.

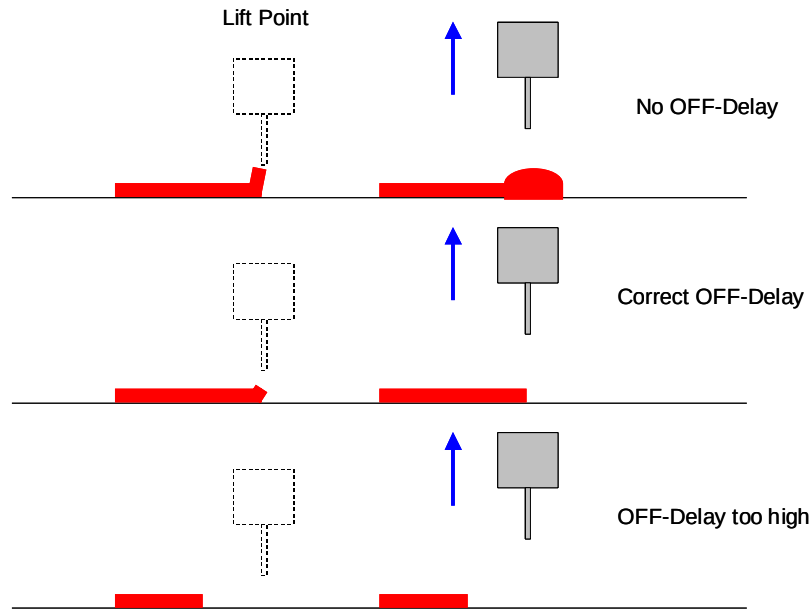


Figure 38. Using and OFF delay to and a dispense line

6.3 Single Part Glob Tops

Single part encapsulants are applied using a spiral in, spiral out or zig-zag dispense pattern. Spiral in is recommended since the perimeter is defined before material is placed inside, so if material runs due to inconsistency in the syringe or some other problem, it is contained. Also because dispensing finishes centrally any spots from the needle are more likely to land onto the component area rather than elsewhere on the PCB. The examples in Figure 39 are for a square, but equally round spirals can be used if desired.

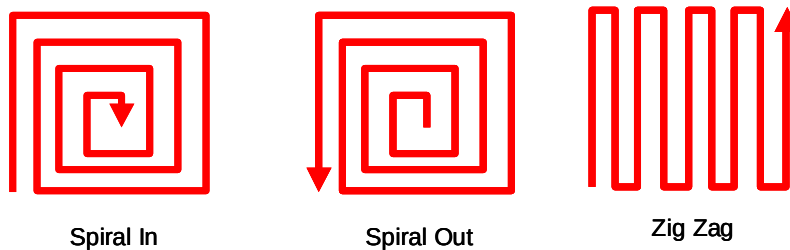


Figure 39. Single part encapsulant dispense patterns

For the spiral pattern you will need to define the overall size, the pitch of the concentric lines, and the speed of the needle. The faster the needle moves the thinner and lower the lines of encapsulant will be. The smaller the pitch the longer the overall spiral will be, meaning more material is deposited. These effects can be seen in Figure 40, which is a photograph of a matrix of *GlobTops* deposited with varying pitch and needle speed.

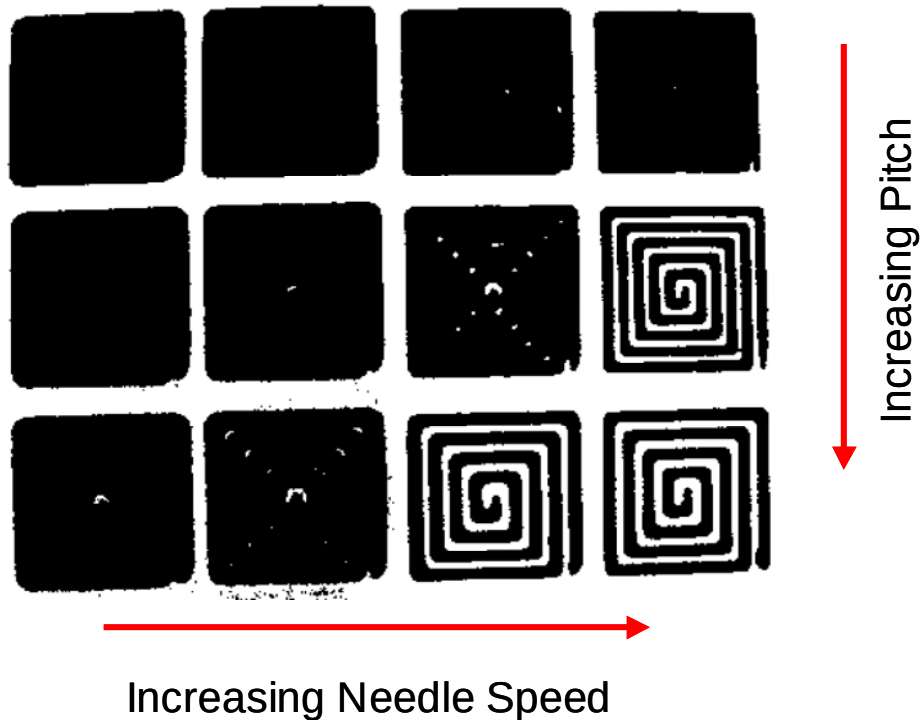


Figure 40. Photograph of GlobTop deposits with varying needle speed at pitch

To determine the correct dispense pattern:

- Firstly one must ensure the height of the dispensed line is high enough to cover the chip component (reduce needle speed to increase height). ?????
- The next step is to ensure enough material overall is applied to avoid voids and gaps in the dispensed pattern (reduce pitch to eliminate gaps).
- Alter ON and OFF delays to prevent pooling at finish or a missing corner at the start point. (See 6.1 and 6.2)

Due to the high viscosity nature of single part GlobTop materials often the spiral dispense pattern will be visible after dispense and cure. A profile of a dispensed GlobTop is shown in Figure 41. These lines can be reduced by heating the glob top material in the nozzle, but frequently these can never be totally eliminated and are entirely acceptable.

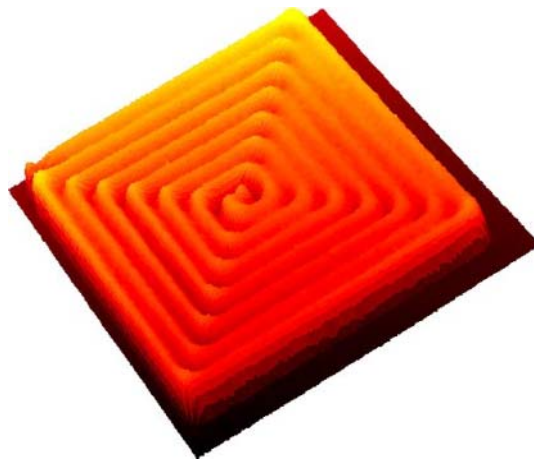


Figure 41. Laser profile showing spiral indentations in cured single part GlobTop

6.4 Dam & Fill GlobTops

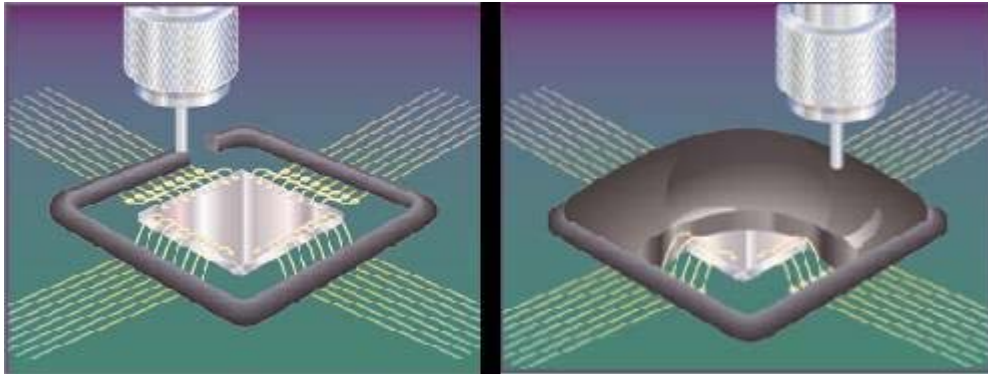


Figure 42. Schematic of a dam and fill process¹

6.4.1 Damming

Dam materials are always high viscosity, and when dispensed will provide a tall barrier, which can later be used to contain the low viscosity fill material.

The height of the dam must of course be sufficient for the fill material to cover the chip component being encapsulated. To increase the height of the dam, reduce the speed of the needle during dispensing. The height can only be increased to a certain point, however, and then the dam material will begin to spread, this should be avoided. Figure 43 shows laser profiles of dam walls. The increasing height and width of a the dam is caused by reducing needle speed. It is useful to note the red line represents an optimum dam condition (2.5mm wide, 0.9mm high) for this material.

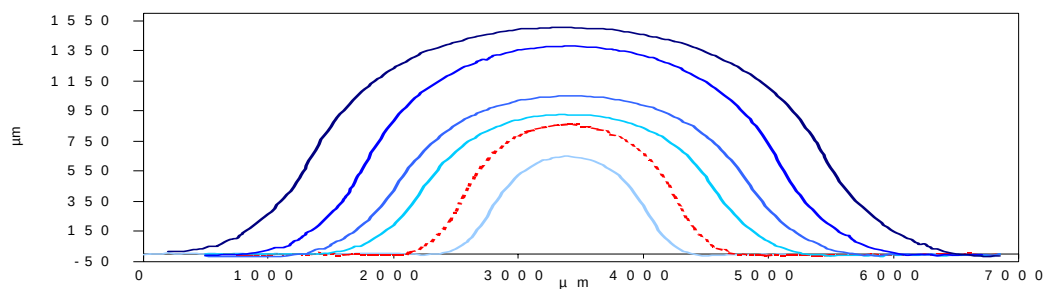


Figure 43. Effect of decreasing needle speed on dam profile

???? axis labels, needle size, dispense speeds

The correct setting of the *ON-Delay* and *OFF-Delay* parameter are critical for a good dam deposit (Figure 37 & Figure 38). There must be a good sealing of the dam at the start/finish point, but without too much pooling. Figure 44 shows the effects of these settings on a dam deposit (no fill) for a sample dam material, the correct parameters produced the square highlighted in red.

???? make red square more bold

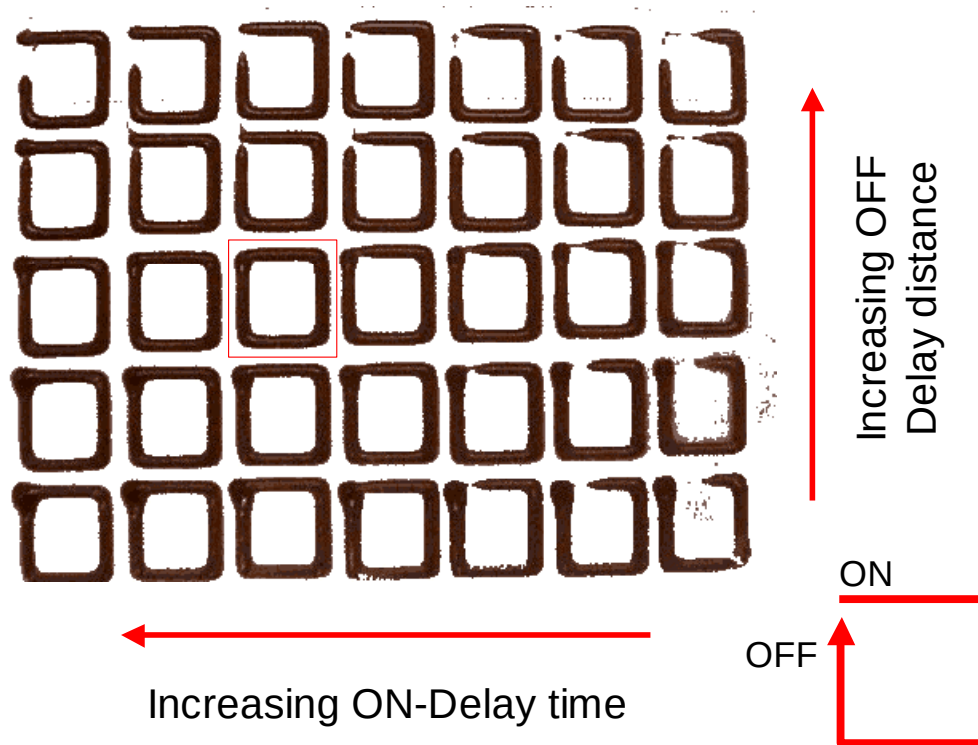


Figure 44. Photograph of dam deposits with varying ON and OFF-Delays

6.4.2 Filling

Self levelling filling materials are designed to be of low viscosity, and have a low surface tension to make them very searching. They should be applied using a spiral-in pattern, with the needle speed and spiral pitch being adjusted to provide enough material to cover the component and reach the dam height. These adjustments are very similar to those used for single part globtops (See section 6.3)

Some fill materials are designed to produce more of a 'glob' shaped curve, rather than a flat levelled encapsulation. These materials are often more viscous and may need to be heated in the nozzle, and can have a higher surface tension to help create a domed shape. They are useful for covering Chip on Board (COB) components, which may have high wire bonds (See Figure 45).

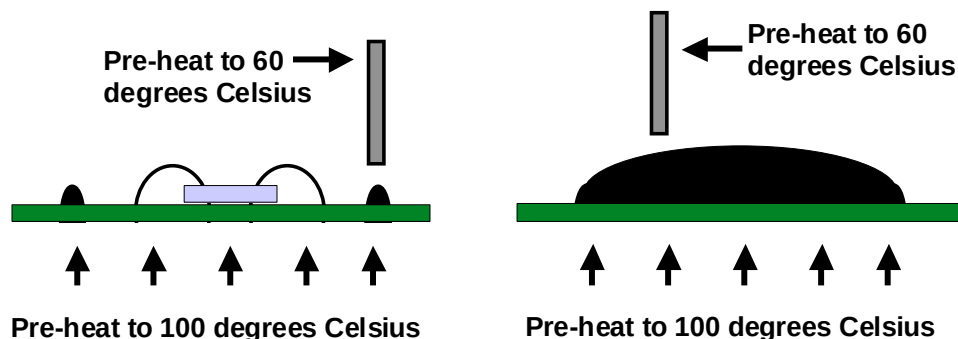


Figure 45. High viscosity fill material used cover COB wire bonds¹

6.5 Heating

Follow the product manufacturers guidelines for specific glob top products. Heating of the laminate to around 100°C and heating of the material in the needle, are generally recommended. The heat helps to allow flow of the globtop under and around the components and helps to avoid air entrapment.

7 DISPENSING UNDERFILLS

Soldered Flip-Chip devices can be underfilled with an encapsulant material. Underfill materials protect the downward facing bare die from the environment, and alleviate the TCE mismatch between the die and the substrate. Using underfill materials increases solder joint reliability and enables larger dies to be used.

Underfills are low in viscosity and have a low surface tension. When applied at the perimeter of a chip they are required to travel beneath the component by capillarity forces, and fill the volume beneath. Modern underfills will produce a fillet all the way around the chip even if originally applied on only one side.

To ensure the correct TCE materials are filled with particles of quartz.

Underfills are also used for larger BGA style components to increase resistance to mechanical shock, driven by more demanding mobile telecom products.

7.1 Dispense Patterns

There are several possibilities for the dispense pattern employed to underfill a component, the most common are shown in Figure 46. The dispensed material is shown in red, and the spreading front and direction are shown in blue. ????? blue lines

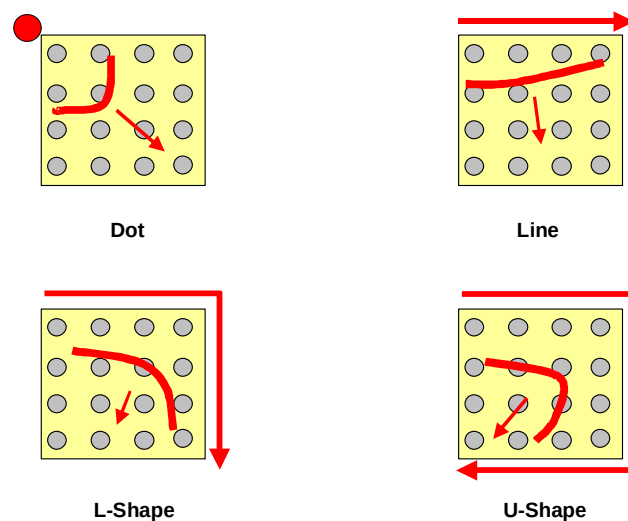


Figure 46. Underfill dispense patterns

- **Dot** – Really only used if needle access to the component is very restricted. Pooling and spreading of material is likely unless material volume is applied in separate lots with a time delay
- **Line** – Useful for components where access may be restricted on other sides
- **L-Shape** – Recommended for most applications. Provides faster fill times than Dot or Line patterns, and is only prone to air entrapment (See 7.2) for some peripheral arrays.
- **U-Shape** – Fastest fill times, but can produce air entrapment (See 7.2).

For larger components the volume of material can be applied in more than one pass to prevent spreading of material from the component edge.

7.2 Air Entrapment and Array Designs

Underfill materials can trap air beneath components if the material travels faster around the perimeter of the component, and closes the air in before material has travelled beneath the chip to displace it. Devices with peripheral bumps are more prone to this, as the bumps aid filling by surface tension, shown schematically in Figure 47.

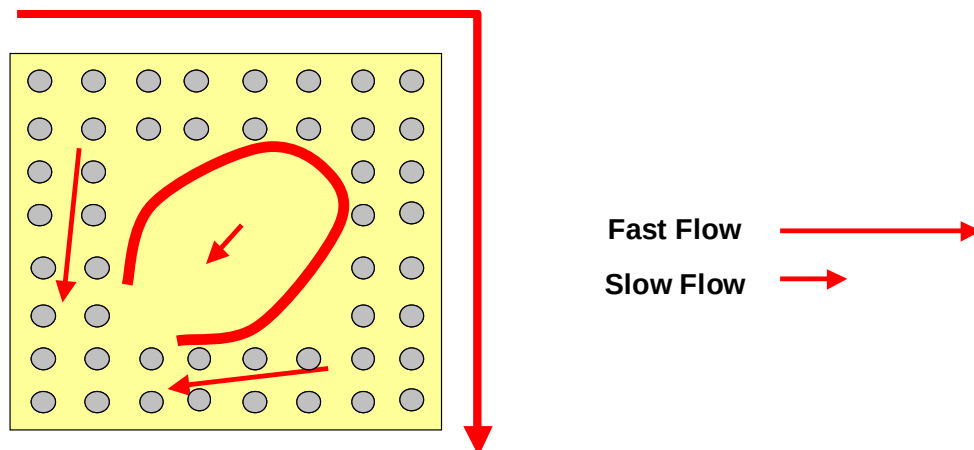


Figure 47. Schematic of fast peripheral flow

???? blue lines

Using a glass slide to model a flip chip component the air trapped beneath can be seen, an example is given in Figure 48. You can reduce the likelihood of air entrapment for peripheral devices by changing the dispense pattern:

- The tendency to trap air: U-Shape > L Shape > Line > Dot.

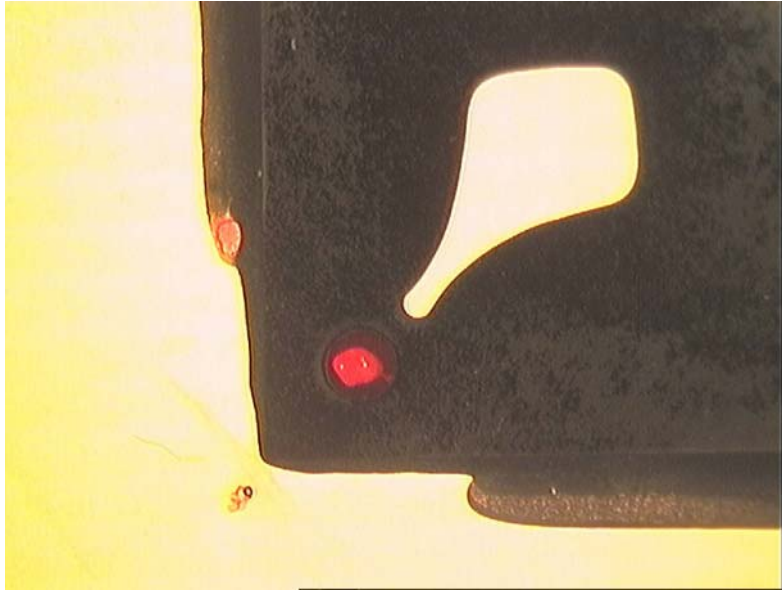


Figure 48. Underfill beneath glass side with air entrapment due to fast peripheral flow

7.3 Underfill Volume

There are many underfill calculating programs available from the web that will help calculate the volume of material you should dispense².

You will need to know:

- Size of the chip
- Gap (or bump size)
- Diameter of bumps
- Tolerances for the above
- Number of bumps
- Desired fillet size (normally at least 1mm)

7.4 Underfill Time

The time taken for underfill materials to flow beneath a component can be critical to the throughput possible in production. Slow underfilling can mean multiple passes are necessary to avoid pooling of material, and delays in beginning the cure of the underfill and further processing.

Figure 49 clearly shows the effects of choice of dispense pattern on fill time:

- Fill Time: U-Shape < L Shape < Line < Dot.

Also highlighted is the increase in fill time for smaller stand off components. Although the force from surface tension is greater at smaller gaps, the quartz filler means the flow is hindered by smaller gaps.

Make sure to specify to your material supplier the component geometries you will be using their product on.

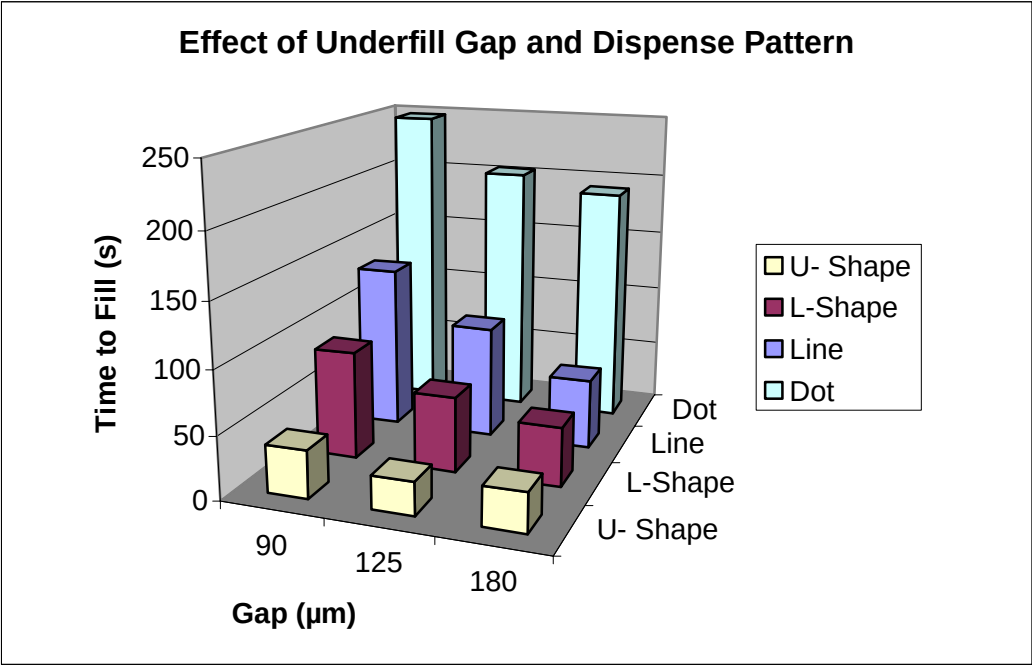


Figure 49. Effect of gap and dispense pattern on fill time (sample underfill on 83°C FR4)

7.5 Heating

Heating of the substrate is critical for underfill operations in terms of the time for the underfill to flow under the component. The recommended substrate temperature is usually from 80°C upwards. Figure 50 shows the effects of substrate temperature on fill times, which increase rapidly to unacceptable levels as the temperature drops.

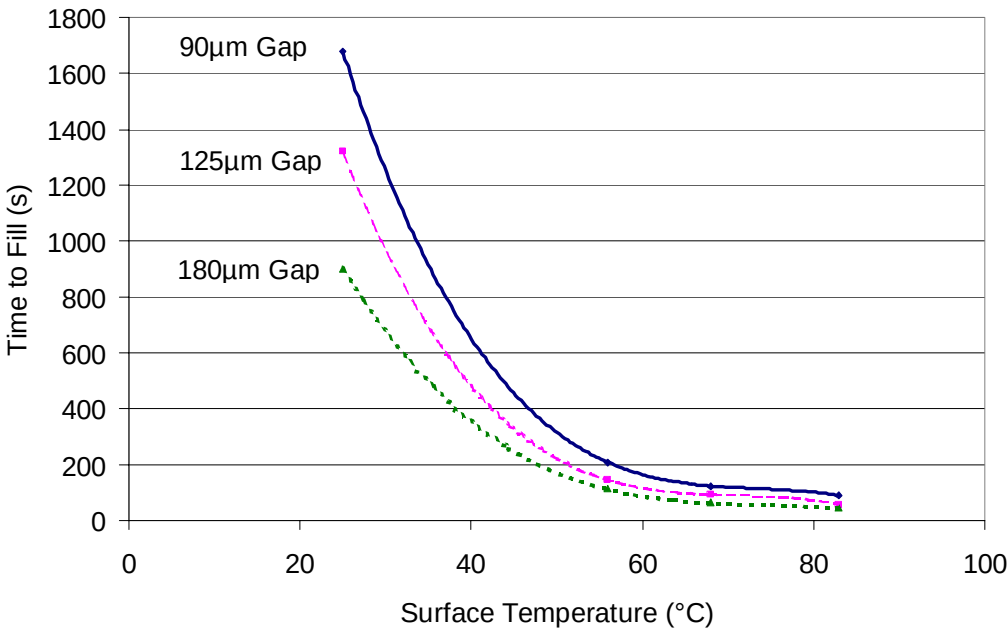


Figure 50. Effect of substrate temperature on fill time, for sample underfill, 22mm square glass slide

If a contact plate can be used to heat the PCB this is preferable, giving even and rapid heating (See section 2.9.2) Contact plates must be tooled for individual products. However, in practice medium to low volume producers need more flexibility and must use non-contact plates heating by radiation and convection. Pre-heaters are normally necessary in these cases.

The ideal temperature for a nozzle assembly for underfill dispensing is around 30°C. Be aware that in some cases needle cooling, as well as heating may be required due to heat from the board area.

8 ACKNOWLEDGMENTS

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 Intertronics
 Loctite - Dr Barry Burns, Kevin Curan and Dr Abayomi Olusanya
 OMG Americas – Chris Palin

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3. NPL Report CMMT(A) 297 - Conductive Adhesives Versus Lead Based Solders: A Comparison Of Performance - *M. Dušek, R Jonck, B. Subotic, J Nottay and P.E. Tomlins*